SILTHAUL · DESIGN DECISION RECORD

SiltHaul R9 accepted with the rear seats folded

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DOCUMENT CONTROL

Document ID	SLH-DDR-004	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/silthaul		
Source	docs/decisions/0004-r9-rear-seats-folded.md @ 9347c2b		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Amish's round-2 decision 2A on R9 recorded

0004: R9 accepted with the rear seats folded

- **Date:** 2026-10-03
- **Status:** decided by Amish, 2026-10-03 (round 2): "i agree with all the 46 recommendations you provided. please proceed." For SiltHaul this is decision 2A, the open decision in SLH-DEC-001 v0.2.

Context

With the bolted flat-pack frame (SLH-DDR-003) the kit packs into about 454 L, 77 % of a small hatchback's load space with the rear seats folded (1,250 x 950 x 500 mm, 594 L). With the seats up, the 890 x 814 mm side frames do not lie flat in a typical 950 x 650 mm boot floor.

Options considered

Option	What it means
A (chosen)	Accept the rear seats folded as the reference; record R9 as met on paper
B	Split each side frame into bolted tubes so everything fits with the seats up: about forty bolts and well over 20 min of setup, which breaks R8

Decision

Option A. R9 is restated as "fits a small hatchback with the rear seats folded; heaviest part 25 kg or less". No design change: the frame, the parts and the mass are as in SLH-DDR-003.

Table 1. Effects.

Item	Result
R9	Met on paper: heaviest lift 23.1 kg; kit about 454 L packed against 594 L
Other requirements	None changed; R8 stays about 20 min, at its limit
Cost	Unchanged. Value-engineering target: USD 2,000. Estimated cost of the constructable design: USD 1,055.60 (USD 944.40 under the target)
Documents	Requirements v0.5, calculation note v0.4 (wording only), build plan v0.3 (pack check wording), register v0.3

Safety

Safety: SiltHaul is a rope-and-anchor system under tension with a hand-cranked chain drive; proof-load tests come before use. Packing the heavy parts into a car is a manual-handling task: no lift is heavier than 25 kg, and two people share the drum with bearings (23.1 kg). Flood mud may carry sewage and disease; use boots, gloves and hygiene measures.